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|---------------|---------------|------------------|
| Surname       | Centre Number | Candidate Number |
| First name(s) |               | 0                |



**GCSE**

**3445U30-1**



**FRIDAY, 10 MAY 2024 – MORNING**

**APPLIED SCIENCE (Double Award)**

**UNIT 3: Food, Materials and Processes**

**FOUNDATION TIER**

**1 hour 30 minutes**

| For Examiner's use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 5            |              |
| 2.                      | 10           |              |
| 3.                      | 11           |              |
| 4.                      | 12           |              |
| 5.                      | 12           |              |
| 6.                      | 6            |              |
| 7.                      | 9            |              |
| 8.                      | 10           |              |
| Total                   | 75           |              |

### ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator, a pencil and a ruler.

### INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

### INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

The Periodic Table is printed on the back cover of the examination paper.



**JUN243445U30101**

Answer **all** questions.

1. Supermarkets need to ensure that food products are safe to eat after purchase. They use different treatments for different foods.

- (a) (i) Join each treatment below to the way it works with a straight line. [2]

| Treatment | How it works                                                       |
|-----------|--------------------------------------------------------------------|
| pickling  | lowers the temperature to stop bacteria multiplying                |
| freezing  | lowers the pH to stop bacteria multiplying                         |
| drying    | removes water so bacteria cannot digest and absorb the food source |

- (ii) Circle **one** other treatment method used to preserve food. [1]

adding oxygen

adding salt

adding water

- (b) Tick (✓) the boxes next to the **two** correct statements about precautions supermarkets take when dealing with raw and cooked meat. [2]

Keep raw and cooked meat apart.

☐

Use the same weighing scales for raw and cooked meat.

☐

Wash hands after handling raw meat.

☐

Keep raw and cooked meat at room temperature.

☐


2. Brewing is an important industry in the UK.

- (a) Complete the word equation for the fermentation of beer by underlining the correct word(s) in the brackets. [1]



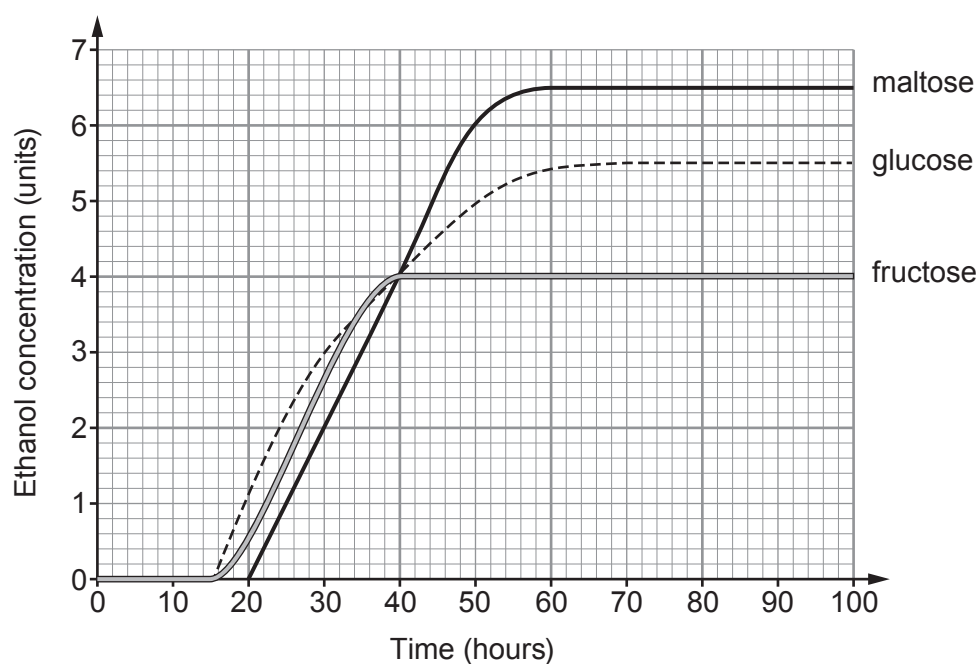
- (b) The main stages of the beer making process are shown below. However, the stages are not in the correct order. [2]

|          |                                                                                          |
|----------|------------------------------------------------------------------------------------------|
| <b>A</b> | starch in the barley is converted to sugar, and water is added to make a brewing mixture |
| <b>B</b> | fermentation stops when all the sugar runs out                                           |
| <b>C</b> | mixture is cooled and yeast is added                                                     |
| <b>D</b> | mixture is boiled                                                                        |
| <b>E</b> | the beer is pasteurised, then stored in bottles or cans                                  |
| <b>F</b> | hops are added for flavour                                                               |

Put the statements in the correct order.

|          |          |  |  |  |          |
|----------|----------|--|--|--|----------|
| <b>A</b> | <b>F</b> |  |  |  | <b>E</b> |
|----------|----------|--|--|--|----------|

- (c) A brewery tested different types of sugar in the fermentation process. These were glucose, maltose and fructose. The results are shown in the graph below.



Use information from the graph and the numbers in the box to answer the following questions.

|   |   |   |    |    |    |
|---|---|---|----|----|----|
| 2 | 3 | 6 | 40 | 50 | 70 |
|---|---|---|----|----|----|

- (i) State the concentration of ethanol formed from **glucose** at **30 hours**. [1]

..... units

- (ii) State the time at which the concentration of the **ethanol** was the same for all the sugars. [1]

time = ..... hours

- (iii) State the time at which fermentation stops for **glucose**. [1]

time = ..... hours

- (iv) Calculate the difference in ethanol concentration between 20 and 50 hours for maltose. [1]

concentration difference = ..... units

- (d) (i) Use your answer in (c)(iv) above and the equation:

$$\text{mean rate} = \frac{\text{concentration difference}}{\text{time difference}}$$

to calculate the mean rate of ethanol formation for **maltose** between 20 and 50 hours. [2]

mean rate = ..... units/hour

- (ii) Charlie says that all sugars produce beer with the same ethanol concentration. State whether you agree **and** give **one** reason for your answer. [1]

.....  
.....



3. Magnesium alloys are used in the casings of tablet computers and mobile phones.

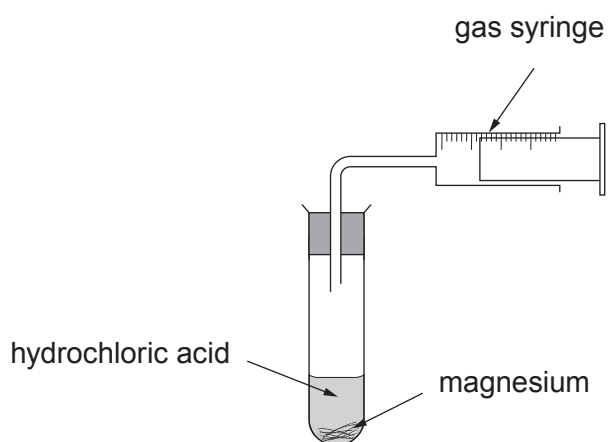
- (a) Complete the sentence below by underlining the correct word in the brackets. [2]

An alloy is a ( **mixture** / **compound** / **solution** ) of two or more elements. One of these must be a ( **gas** / **metal** / **liquid** ).

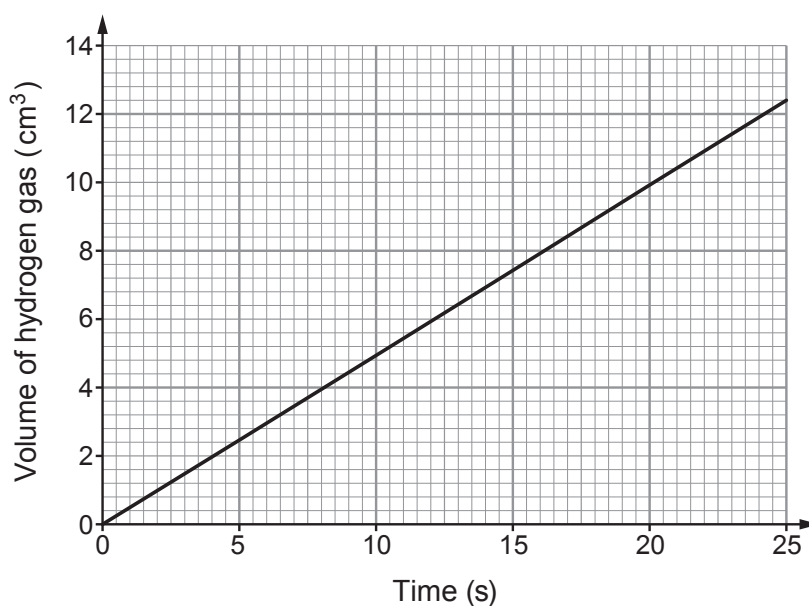
- (b) Magnesium reacts with hydrochloric acid producing hydrogen gas.



A student investigated this reaction using the apparatus below.



A graph of the student's results is shown below.



**Draw** a line on the graph above to show the results that you would expect if the experiment was repeated with a lower concentration of hydrochloric acid. [1]



- (c) The student also investigated the reaction of hydrochloric acid and magnesium at different initial temperatures.

| Initial temperature (°C) | Time taken for magnesium to disappear (s) |
|--------------------------|-------------------------------------------|
| 10                       | 80                                        |
| 20                       | 40                                        |
| 30                       | 20                                        |
| 40                       | 10                                        |
| 50                       | 5                                         |

- (i) State **three** variables that need to be controlled in this experiment. [3]

1. ....
2. ....
3. ....

- (ii) I. The student thinks that every time the temperature doubles, the time taken halves. Explain whether you agree. [2]

.....

.....

.....

.....

- II. Predict the time it would take for the magnesium to disappear at 60 °C. [1]

time = ..... s

- (iii) Complete the following sentences by underlining the correct words in the brackets. [2]

As the temperature increases, particles move  
( **faster** / **slower** / **at the same speed** ).

The particles collide ( **more often** / **as often** / **less often** ) in a given time.

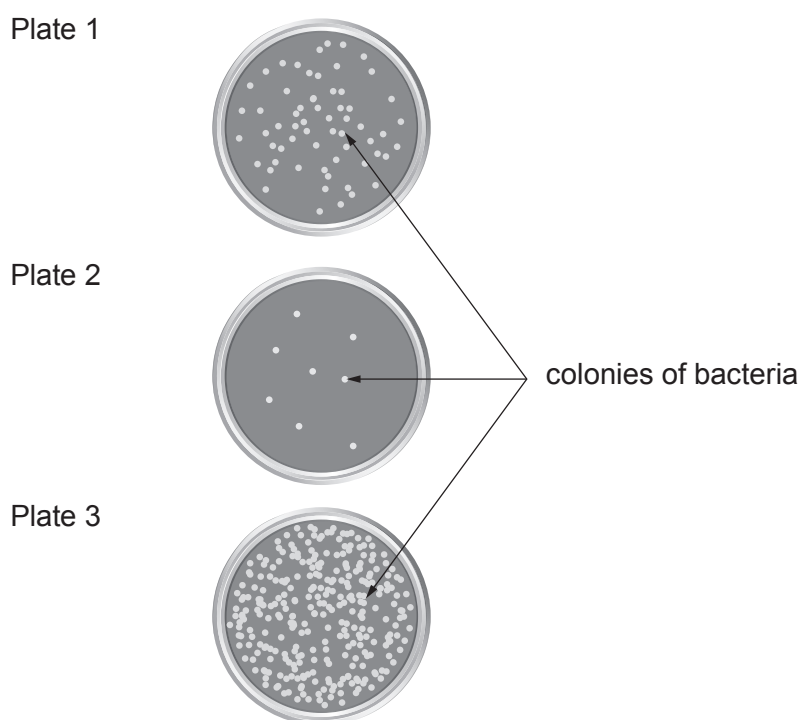


4. A laboratory technician tests **three** different food samples taken from a local restaurant by an Environmental Health Officer.

The samples were spread onto sterile agar plates which were then incubated at 37 °C.

After 3 days the agar plates were examined. The results are shown below.

It is assumed that each colony was formed from one bacterium.



- (a) (i) The technician concludes that the food that produced Plate 1 was stored in the restaurant at 15 °C, the food that produced Plate 2 was stored at 25 °C, and the food that produced Plate 3 was stored at 5 °C. Explain whether you agree. [3]

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- (ii) State why it is not possible to make a conclusion on the reproducibility of this analysis. [1]

.....

.....



(b) Agar plates can also be used in a school laboratory to investigate bacterial growth.

(i) State why the agar plate needs to be sterile. [1]

.....

(ii) State what should be done to the inoculating loop before it is used to transfer the food sample to the agar plate. [1]

.....

(iii) Give a reason why the students are only allowed to incubate their plates at 25 °C. [1]

.....

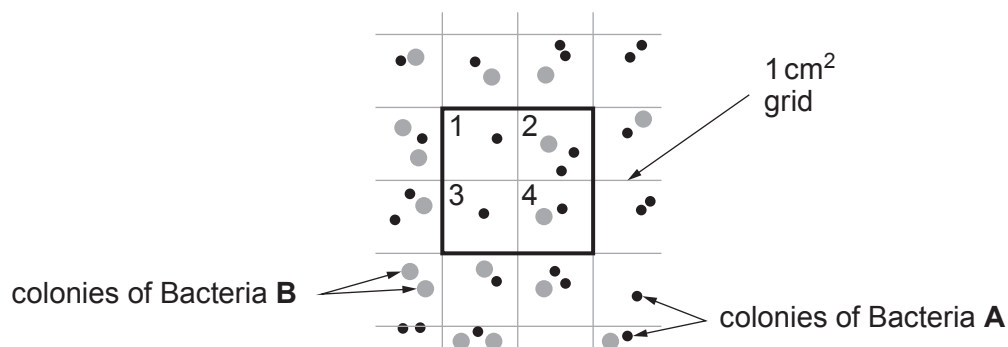
(iv) State why it is important the lid remains on the agar plate during incubation. [1]

.....



- (c) Two types of bacteria, **A** and **B**, were found on one of the plates. The plate has an area of  $52\text{ cm}^2$ . Each square in the grid is  $1\text{ cm}^2$ .

Section of agar plate



The results for four squares numbered **1** to **4** are shown in the table below. The students have completed the table for Bacteria **A**.

| Grid number                              | Number of bacteria in $1\text{ cm}^2$ |                   |
|------------------------------------------|---------------------------------------|-------------------|
|                                          | Bacteria <b>A</b>                     | Bacteria <b>B</b> |
| 1                                        | 1                                     | 0                 |
| 2                                        | 2                                     | 1                 |
| 3                                        | 1                                     | 0                 |
| 4                                        | 1                                     | .....             |
| Mean colonies per $\text{cm}^2$          | 1.25                                  | .....             |
| Mean colonies per $52\text{ cm}^2$ plate | 65                                    | 26                |
| Total number of colonies in squares 1–4  | 5                                     | .....             |

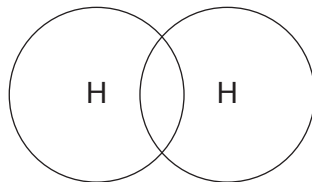
- (i) Complete the table for Bacteria **B**. [3]
- (ii) State how the results would be improved by counting the number of bacteria in 12 squares. [1]

.....



5. Hydrogen could be used to produce hydrogen chloride gas.

- (a) The diagram represents the bonding in hydrogen. Complete the dot and cross diagram for a molecule of hydrogen. [1]



- (b) The reaction between hydrogen and chlorine can be represented by the following equation.



The bond energy values are shown in the table.

| Bond  | Energy needed to break or form the bond (kJ) |
|-------|----------------------------------------------|
| H-H   | 436                                          |
| Cl-Cl | 242                                          |
| H-Cl  | 431                                          |

Use the information in the table to answer the following questions.

- (i) Calculate the energy needed to break all the bonds in the reactants. [2]

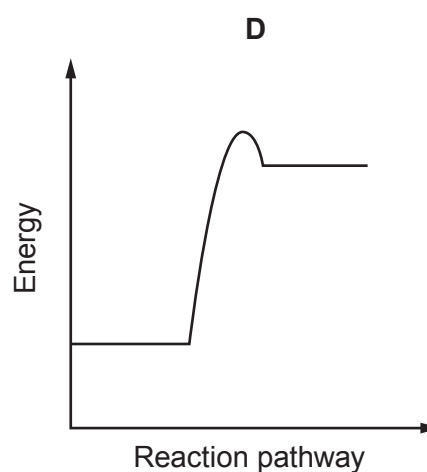
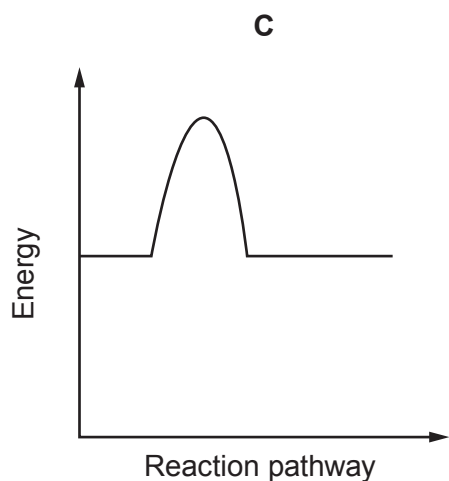
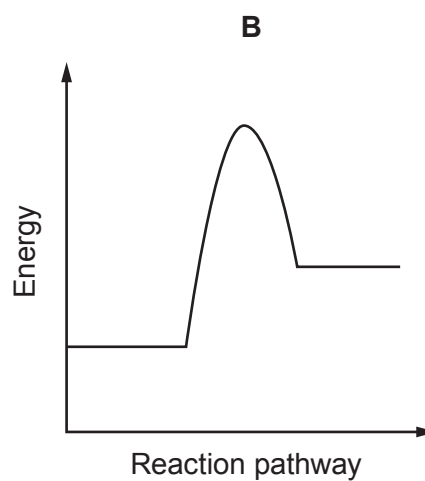
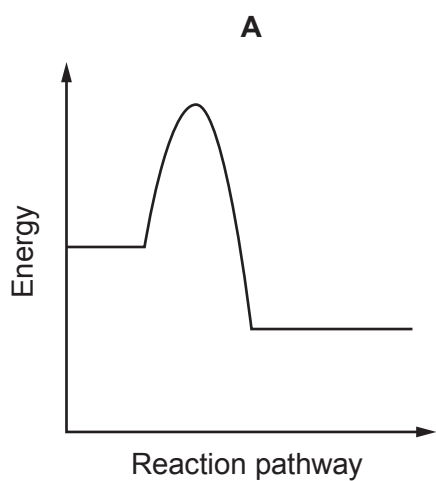
energy = ..... kJ

- (ii) Calculate the energy given out when all the bonds in the product are formed. [2]

energy = ..... kJ



(iii) Possible energy diagrams for this reaction are shown below.



I. Select the correct energy diagram for this reaction. .... [1]

II. Explain your choice. [2]

.....  
.....



- (c) Use words from the box to complete the sentences below about an **endothermic** reaction. [4]

more    less    taken in    given out    increase    decrease    thermal runaway

In an endothermic reaction ..... energy is stored in the bonds **after** the reaction than before. This means energy is ..... from the surroundings.

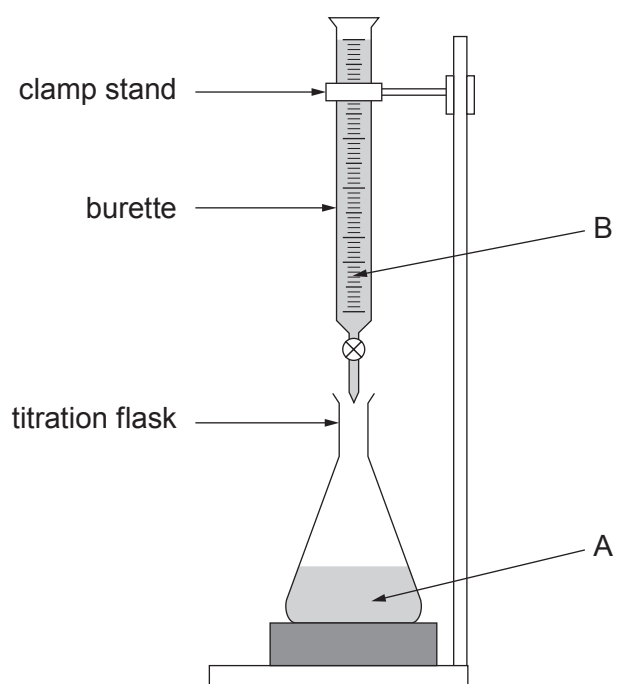
This will cause the temperature to ..... during the reaction.

This cannot lead to a ..... reaction.

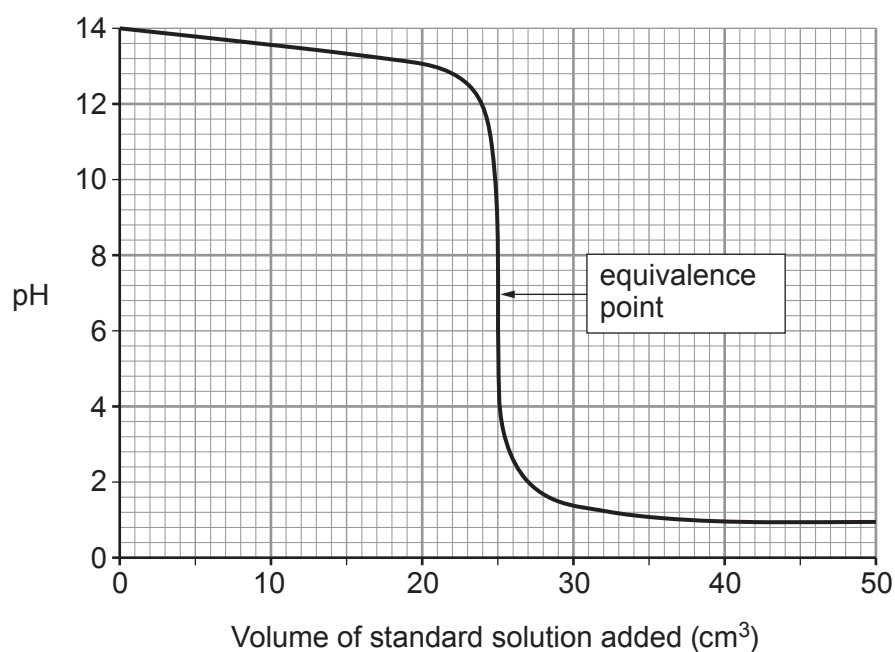


6. Solutions can be acidic, alkaline or neutral. Analytical chemists use titrations to find the concentrations of solutions.

The concentration of solution A can be found by titrating against a standard solution B of known concentration.



The titration curve is shown below.



Explain the **pH** changes during the titration.

Include in your answer:

- what can be deduced from the curve about solutions A and B
- the point at which the solution becomes neutral
- which solution is acid and which solution is alkaline.

[6 QER]

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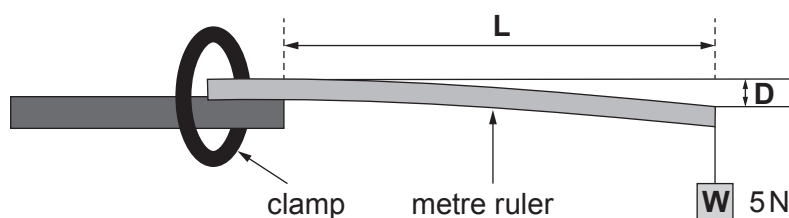
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.....

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7. Students are investigating how the stiffness of a wooden metre ruler depends on its length. They use the apparatus shown below. The weight, **W**, on the end of the ruler remains constant at 5 N.



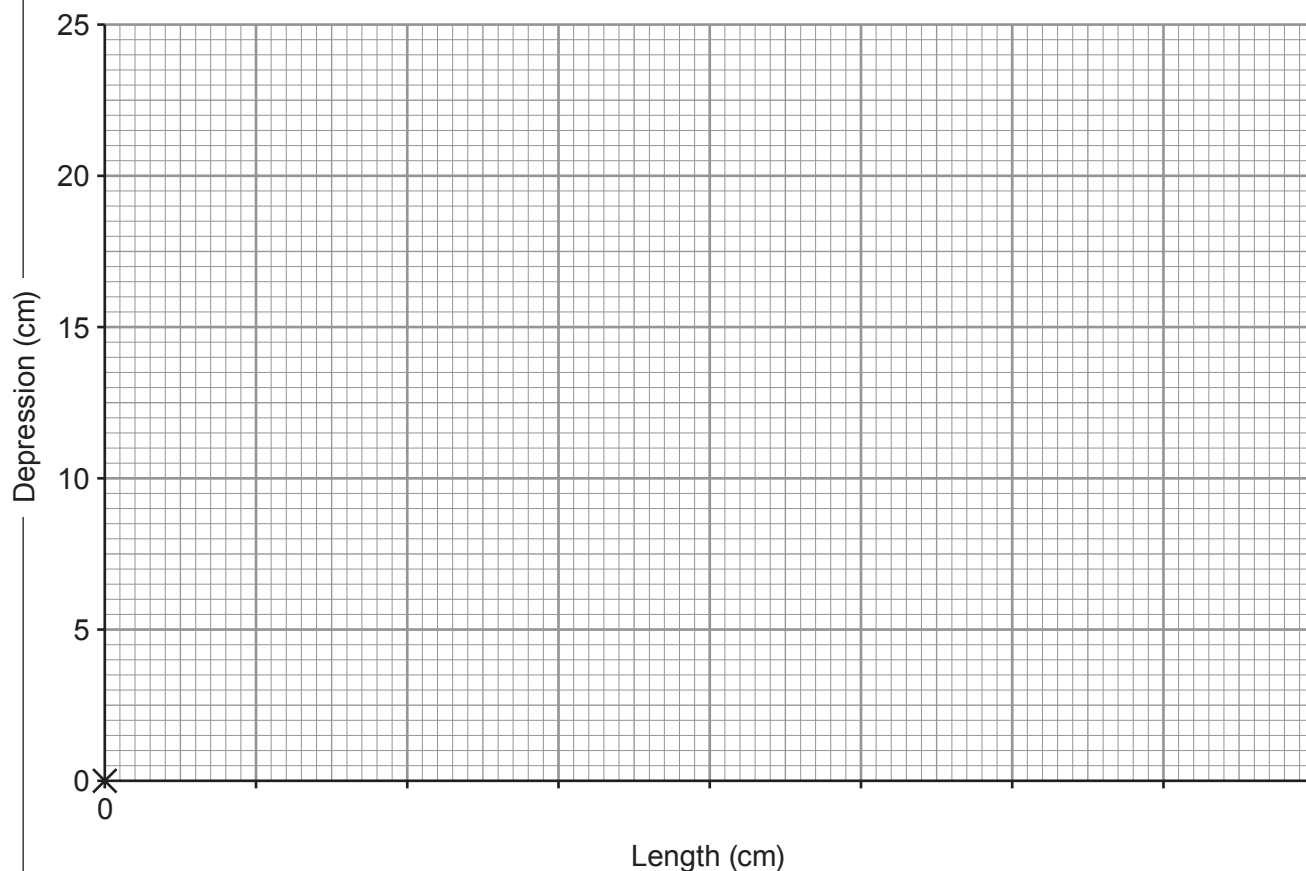
The length, **L**, is varied. For each length, **L**, the depression, **D**, of the end of the ruler is measured.

Their results are shown in the table below.

| Length <b>L</b> (cm) | Depression <b>D</b> (cm) |
|----------------------|--------------------------|
| 0                    | 0.0                      |
| 30                   | 1.0                      |
| 40                   | 3.0                      |
| 50                   | 6.0                      |
| 60                   | 10.0                     |
| 70                   | 15.0                     |
| 80                   | 22.0                     |



- (a) Plot the data on the grid below and draw a smooth curve. The point (0,0) has already been plotted for you. [4]



- (b) It was expected that depression, **D**, would be directly proportional to length, **L**. Explain whether the graph shows a proportional relationship. [2]

.....

.....

.....

- (c) Use information in the table and the equation:

$$\text{stiffness} = \frac{\text{weight}}{\text{depression}}$$

to calculate the stiffness of the ruler at a length of 50 cm. [3]

stiffness = ..... N/cm

9

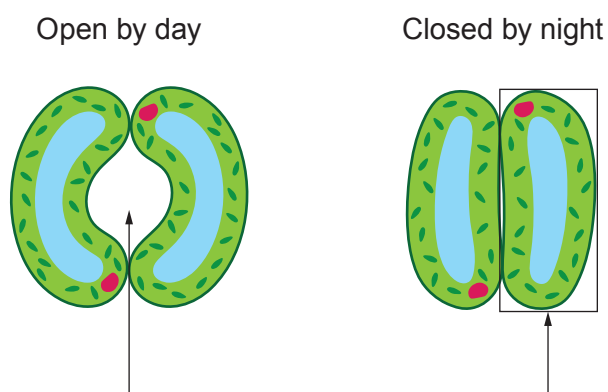


8. Photosynthesis is the process by which plants make food from water and carbon dioxide.

(a) Stomata are an important part of the leaf. In many plants, stomata remain open during the day and closed at night.

(i) Label the diagram below.

[2]



(ii) State **two** functions of stomata.

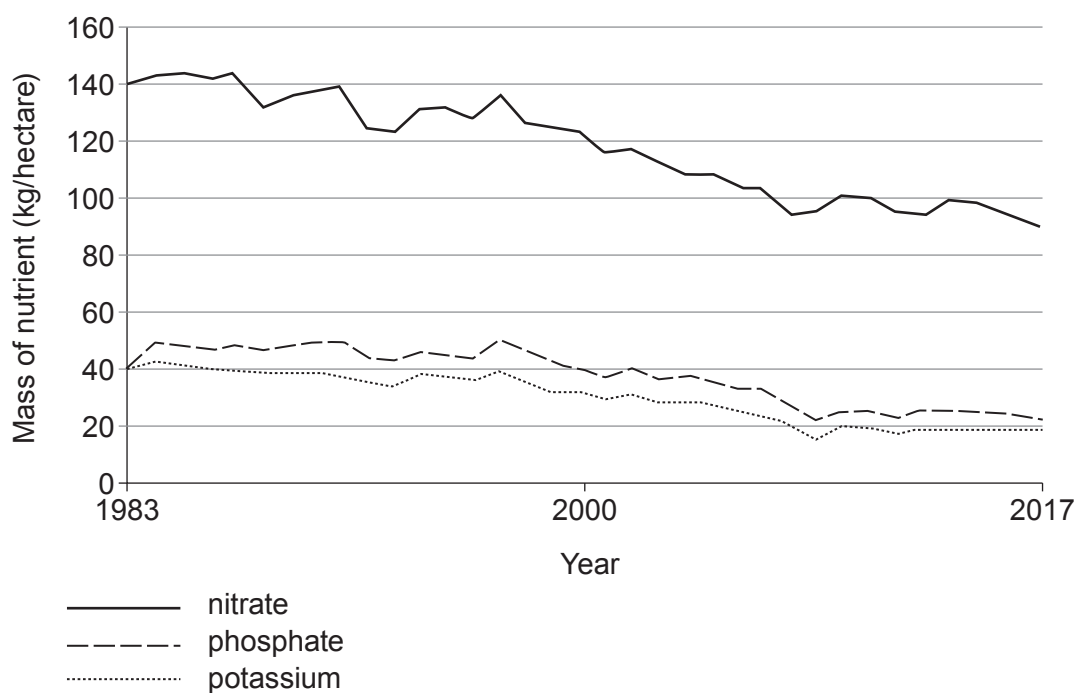
[2]

1. ....
2. ....



- (b) Healthy plants require a supply of nutrients. Many farmers use artificial fertilisers to supply these nutrients to the soil.

The graph shows the mass of different nutrients applied to agricultural land between 1983 and 2017.



Use the data from the graph to answer the following questions.

- (i) It is claimed that the use of nitrate fertilisers halved between 1983 and 2017. Use the graph to explain whether you agree.

[3]

.....

.....

.....



- (ii) The total land area of Wales is approximately 2 100 000 hectares.  
60% of this land is used for farming.

I. Calculate the area of land used for farming in Wales. [1]

area = ..... hectares

II. Calculate the total mass of phosphates used in farming in Wales in 1983. [2]

mass = ..... kg

**END OF PAPER**

|    |
|----|
|    |
| 10 |





# THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

|                                                                                                                                                                                 |  |                                                                                   |  |                                                                                    |  |                                                                                   |  |                                                                                   |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                  |  |                                                                                  |  |                                                                                   |  |                                                                                   |  |                                                                                   |  |                                                                                    |  |                                                                                   |  |                                                                                 |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------|--|
| <div><div><div><div><div>1</div><div>H</div><div>Hydrogen</div><div>1</div></div></div><div><div><div>4</div><div>He</div><div>Helium</div><div>2</div></div></div></div></div> |  |                                                                                   |  |                                                                                    |  |                                                                                   |  |                                                                                   |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                  |  |                                                                                  |  |                                                                                   |  |                                                                                   |  |                                                                                   |  |                                                                                    |  |                                                                                   |  |                                                                                 |  |
| <div><div><div>7</div><div>Li</div><div>Lithium</div><div>3</div></div></div>                                                                                                   |  | <div><div><div>9</div><div>Be</div><div>Beryllium</div><div>4</div></div></div>   |  |                                                                                    |  | <div><div><div>11</div><div>B</div><div>Boron</div><div>5</div></div></div>       |  | <div><div><div>12</div><div>C</div><div>Carbon</div><div>6</div></div></div>      |  | <div><div><div>14</div><div>N</div><div>Nitrogen</div><div>7</div></div></div>     |  | <div><div><div>16</div><div>O</div><div>Oxygen</div><div>8</div></div></div>       |  | <div><div><div>19</div><div>F</div><div>Fluorine</div><div>9</div></div></div>     |  | <div><div><div>20</div><div>Ne</div><div>Neon</div><div>10</div></div></div>       |  |                                                                                    |  |                                                                                  |  |                                                                                  |  |                                                                                   |  |                                                                                   |  |                                                                                   |  |                                                                                    |  |                                                                                   |  |                                                                                 |  |
| <div><div><div>23</div><div>Na</div><div>Sodium</div><div>11</div></div></div>                                                                                                  |  | <div><div><div>24</div><div>Mg</div><div>Magnesium</div><div>12</div></div></div> |  |                                                                                    |  |                                                                                   |  | <div><div><div>27</div><div>Al</div><div>Aluminium</div><div>13</div></div></div> |  | <div><div><div>28</div><div>Si</div><div>Silicon</div><div>14</div></div></div>    |  | <div><div><div>31</div><div>P</div><div>Phosphorus</div><div>15</div></div></div>  |  | <div><div><div>32</div><div>S</div><div>Sulfur</div><div>16</div></div></div>      |  | <div><div><div>35.5</div><div>Cl</div><div>Chlorine</div><div>17</div></div></div> |  | <div><div><div>40</div><div>Ar</div><div>Argon</div><div>18</div></div></div>      |  |                                                                                  |  |                                                                                  |  |                                                                                   |  |                                                                                   |  |                                                                                   |  |                                                                                    |  |                                                                                   |  |                                                                                 |  |
| <div><div><div>39</div><div>K</div><div>Potassium</div><div>19</div></div></div>                                                                                                |  | <div><div><div>40</div><div>Ca</div><div>Calcium</div><div>20</div></div></div>   |  | <div><div><div>45</div><div>Sc</div><div>Scandium</div><div>21</div></div></div>   |  | <div><div><div>48</div><div>Ti</div><div>Titanium</div><div>22</div></div></div>  |  | <div><div><div>51</div><div>V</div><div>Vanadium</div><div>23</div></div></div>   |  | <div><div><div>52</div><div>Cr</div><div>Chromium</div><div>24</div></div></div>   |  | <div><div><div>55</div><div>Mn</div><div>Manganese</div><div>25</div></div></div>  |  | <div><div><div>56</div><div>Fe</div><div>Iron</div><div>26</div></div></div>       |  | <div><div><div>59</div><div>Co</div><div>Cobalt</div><div>27</div></div></div>     |  | <div><div><div>59</div><div>Ni</div><div>Nickel</div><div>28</div></div></div>     |  | <div><div><div>63.5</div><div>Cu</div><div>Copper</div><div>29</div></div></div> |  | <div><div><div>65</div><div>Zn</div><div>Zinc</div><div>30</div></div></div>     |  | <div><div><div>70</div><div>Ga</div><div>Gallium</div><div>31</div></div></div>   |  | <div><div><div>73</div><div>Ge</div><div>Germanium</div><div>32</div></div></div> |  | <div><div><div>75</div><div>As</div><div>Arsenic</div><div>33</div></div></div>   |  | <div><div><div>79</div><div>Se</div><div>Selenium</div><div>34</div></div></div>   |  | <div><div><div>80</div><div>Br</div><div>Bromine</div><div>35</div></div></div>   |  | <div><div><div>84</div><div>Kr</div><div>Krypton</div><div>36</div></div></div> |  |
| <div><div><div>86</div><div>Rb</div><div>Rubidium</div><div>37</div></div></div>                                                                                                |  | <div><div><div>88</div><div>Sr</div><div>Strontium</div><div>38</div></div></div> |  | <div><div><div>89</div><div>Y</div><div>Yttrium</div><div>39</div></div></div>     |  | <div><div><div>91</div><div>Zr</div><div>Zirconium</div><div>40</div></div></div> |  | <div><div><div>93</div><div>Nb</div><div>Niobium</div><div>41</div></div></div>   |  | <div><div><div>96</div><div>Mo</div><div>Molybdenum</div><div>42</div></div></div> |  | <div><div><div>99</div><div>Tc</div><div>Technetium</div><div>43</div></div></div> |  | <div><div><div>101</div><div>Ru</div><div>Ruthenium</div><div>44</div></div></div> |  | <div><div><div>103</div><div>Rh</div><div>Rhodium</div><div>45</div></div></div>   |  | <div><div><div>106</div><div>Pd</div><div>Palladium</div><div>46</div></div></div> |  | <div><div><div>108</div><div>Ag</div><div>Silver</div><div>47</div></div></div>  |  | <div><div><div>112</div><div>Cd</div><div>Cadmium</div><div>48</div></div></div> |  | <div><div><div>115</div><div>In</div><div>Indium</div><div>49</div></div></div>   |  | <div><div><div>119</div><div>Sn</div><div>Tin</div><div>50</div></div></div>      |  | <div><div><div>122</div><div>Sb</div><div>Antimony</div><div>51</div></div></div> |  | <div><div><div>128</div><div>Te</div><div>Tellurium</div><div>52</div></div></div> |  | <div><div><div>127</div><div>I</div><div>Iodine</div><div>53</div></div></div>    |  | <div><div><div>131</div><div>Xe</div><div>Xenon</div><div>54</div></div></div>  |  |
| <div><div><div>133</div><div>Cs</div><div>Caesium</div><div>55</div></div></div>                                                                                                |  | <div><div><div>137</div><div>Ba</div><div>Barium</div><div>56</div></div></div>   |  | <div><div><div>139</div><div>La</div><div>Lanthanum</div><div>57</div></div></div> |  | <div><div><div>179</div><div>Hf</div><div>Hafnium</div><div>72</div></div></div>  |  | <div><div><div>181</div><div>Ta</div><div>Tantalum</div><div>73</div></div></div> |  | <div><div><div>184</div><div>W</div><div>Tungsten</div><div>74</div></div></div>   |  | <div><div><div>186</div><div>Re</div><div>Rhenium</div><div>75</div></div></div>   |  | <div><div><div>190</div><div>Os</div><div>Osmium</div><div>76</div></div></div>    |  | <div><div><div>192</div><div>Ir</div><div>Iridium</div><div>77</div></div></div>   |  | <div><div><div>195</div><div>Pt</div><div>Platinum</div><div>78</div></div></div>  |  | <div><div><div>197</div><div>Au</div><div>Gold</div><div>79</div></div></div>    |  | <div><div><div>201</div><div>Hg</div><div>Mercury</div><div>80</div></div></div> |  | <div><div><div>204</div><div>Tl</div><div>Thallium</div><div>81</div></div></div> |  | <div><div><div>207</div><div>Pb</div><div>Lead</div><div>82</div></div></div>     |  | <div><div><div>209</div><div>Bi</div><div>Bismuth</div><div>83</div></div></div>  |  | <div><div><div>210</div><div>Po</div><div>Polonium</div><div>84</div></div></div>  |  | <div><div><div>210</div><div>At</div><div>Astatine</div><div>85</div></div></div> |  | <div><div><div>222</div><div>Rn</div><div>Radon</div><div>86</div></div></div>  |  |
| <div><div><div>223</div><div>Fr</div><div>Francium</div><div>87</div></div></div>                                                                                               |  | <div><div><div>226</div><div>Ra</div><div>Radium</div><div>88</div></div></div>   |  | <div><div><div>227</div><div>Ac</div><div>Actinium</div><div>89</div></div></div>  |  |                                                                                   |  |                                                                                   |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                    |  |                                                                                  |  |                                                                                  |  |                                                                                   |  |                                                                                   |  |                                                                                   |  |                                                                                    |  |                                                                                   |  |                                                                                 |  |

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